ch22 AC and Domestic Electricity

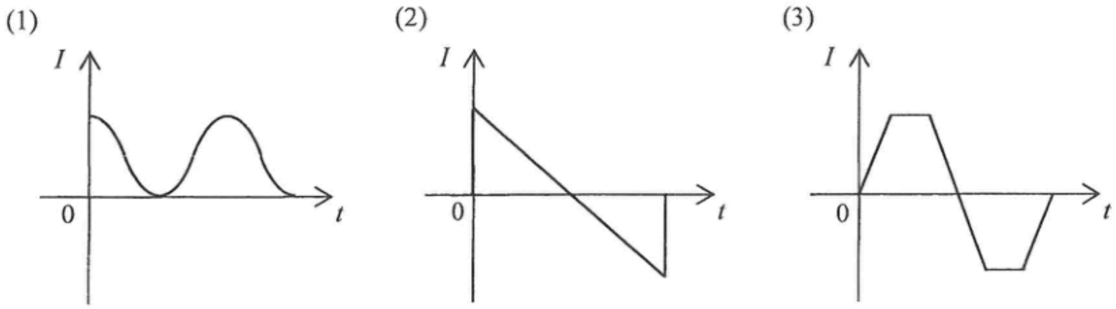
2016 Q28

28. A television set in stand-by mode consumes 1.5 W. If it is in this mode for 16 hours a day, estimate the carbon dioxide (CO_2) emission due to the electricity consumed in stand-by mode in a 30-day month. Given: 1 kW h of electricity consumed corresponds to 0.8 kg CO_2 emission from the power station.

A. 0.576 kg
B. 0.720 kg
C. 576 kg
D. 720 kg

2024 Q28

*28. Which of the following current against time (I- t) graphs represent an a.c. ?



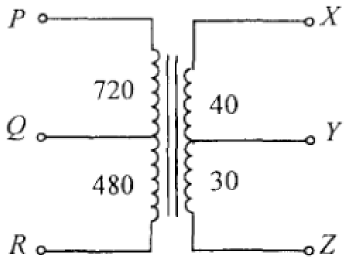
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2017 Q25

25. Which of the following statements about the use of a fuse is correct ?

A. A fuse should be installed in the neutral wire.
B. A fuse is not required in an electrical appliance with double insulation.
C. A 5A fuse is suitable for a heater of rating '220 V, 1500 W'.
D. The melting point of a fuse should be lower than that of copper.

*30.



The above figure represents a multi-tapped transformer. The number of turns between the various 'tapping points' are indicated as shown. Which connections should be used for stepping down a voltage from 240 V to 6 V ?

	primary coil	**secondary coil**
A.	*PQ*	*XY*
B.	*PQ*	*YZ*
C.	*PR*	*XY*
D.	*PR*	*YZ*

2017 Q29

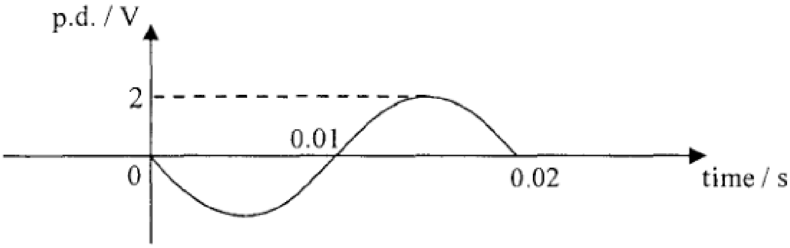
*29. A heater of resistance 100 Ω is connected to the mains supply. The r.m.s. voltage of the mains supply is 110 V. Which of the following statements are correct ?

(1) The peak voltage across the heater is 156 V.
(2) The power dissipated by the heater is 121 W.
(3) The power dissipated by the heater will be doubled if the r.m.s. voltage of the mains supply doubles.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2021 Q28

*28. The graph shows the waveform of a sinusoidal alternating p.d. applied across a 10 Ω resistor.



Find the root-mean-square current in this 10 Ω resistor and the average power dissipated by it.

	root-mean-square current / A	average power / W
A.	0.14	0.2
B.	0.14	0.4
C.	0.2	0.2
D.	0.2	0.4

2015 Q28

28. Which statement is **NOT** a reason why mains sockets at home are connected in parallel instead of a series circuit ?

A. Electrical appliances connected to different sockets can be switched on or off independently.

B. Voltage supplied to each socket is fixed and all electrical appliances can operate at their rated voltages.

C. The current supplied can be reduced and thinner cables can then be used.

D. When an electrical appliance breaks down and becomes an open circuit, other appliances can still work normally.

2015 Q29

29. An electric iron of 1800 W sold in Hong Kong (220 V 50 Hz) is connected to a 110 V 60 Hz mains socket in another country. How does its performance compare on the same ironing setting ?

A. The electric iron does not work because the a.c. supply is 60 Hz instead of 50 Hz.
B. The electric iron is as hot as when it is used in Hong Kong.
C. The electric iron is hotter than when it is used in Hong Kong.
D. The electric iron is colder than when it is used in Hong Kong.

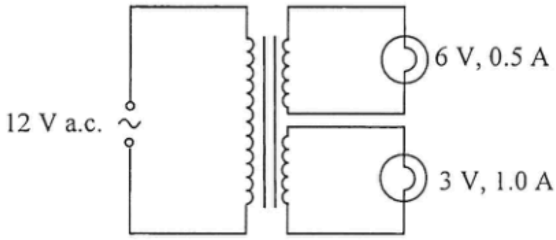
2014 Q30

*30. When a heater is connected to a d.c. voltage of 10 V, the power dissipated is P. If the heater is connected to a sinusoidal a.c., the power dissipated becomes $\frac{1}{2}P$. What is the **r.m.s. voltage** of this a.c. source ? Assume that the resistance of the heater is constant.

A. 5 V
B. $5\sqrt{2}$ V
C. 10 V
D. $10\sqrt{2}$ V

2020 Q29

*29.

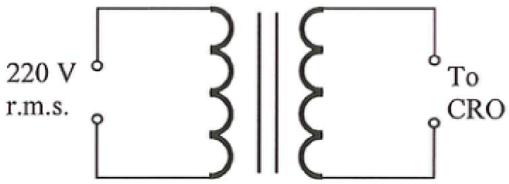


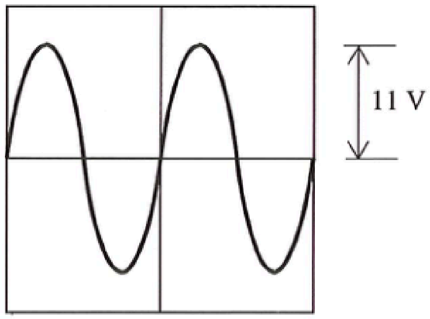
The figure shows an ideal transformer with two secondary coils connected to two light bulbs marked '6 V, 0.5 A' and '3 V, 1.0 A' respectively. When a 12 V a.c. supply is connected to the primary coil, the bulbs work at their respective rated values. Estimate the current in the primary coil.

A. 0.25 A
B. 0.50 A
C. 0.75 A
D. 1.0 A

2023 Q29

*29. A transformer is connected to a.c. mains of 220 V r.m.s. value. The peak value of the output voltage displayed on a CRO is found to be 11 V. If the transformer has 2000 turns in its primary coil, find the number of turns in the secondary coil.

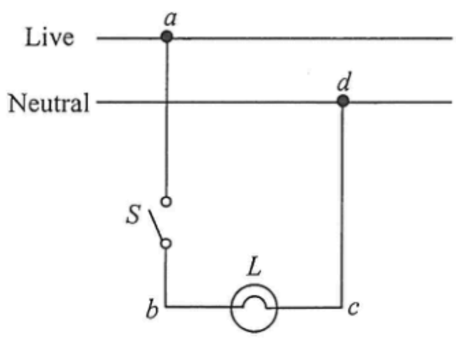




A. 71
B. 100
C. 141
D. 200

2020 Q24

24. The figure shows part of a domestic lighting circuit in which the bulb L does not light up when switch S is closed.



The circuit is then checked with switch S closed. Using a voltage tester to touch points b and c in turns, the tester indicates that both points are at high voltage. When touching points a and d in turns, the tester indicates only point a is at high voltage. Which of the following can be a reason for the fault ?

A. The switch S has been damaged.
B. The filament of bulb L has been burnt out and becomes an open circuit.
C. There is a short circuit between a and d.
D. There is an open circuit between c and d.

2013 Q33

33. Which of the following domestic electrical appliances consumes a power close to 1 kW in normal working conditions ?

A. an electric fan
B. a microwave oven
C. a fluorescent lamp
D. a TV set

2025 Q30

*30. An electric iron works at its rated power of 1100 W under a 50 Hz a.c. mains of root-mean-square voltage 220 V. What power will it deliver when working under a 100 Hz a.c. of root-mean-square voltage 110 V ?

A. 275 W
B. 778 W
C. 550 W
D. 1100 W

2019 Q30

*30. The power consumption of the heating element of an electric heater connected to an a.c. mains can be increased by

(1) increasing the electrical resistance of the heating element.
(2) increasing the frequency of the a.c. voltage.
(3) increasing the r.m.s. value of the a.c. voltage.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2022 Q30

*30. Arrange the following in ascending order of power delivered when each of them is connected to the same resistor.

(1) a 100 Hz sinusoidal a.c. of peak voltage 2 V
(2) a 50 Hz sinusoidal a.c. of r.m.s. voltage 2 V
(3) a steady d.c. of voltage 1.5 V

A. (1) (3) (2)
B. (2) (3) (1)
C. (1) (2) (3)
D. (2) (1) (3)

2024 Q31

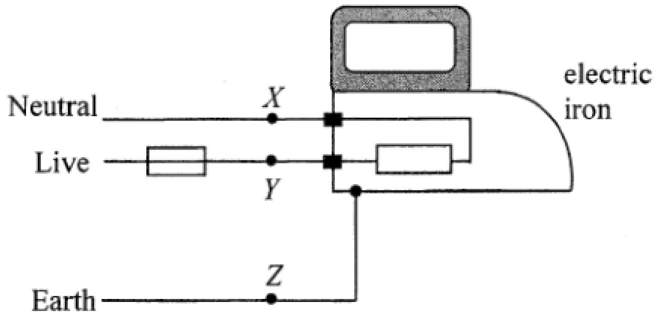
31. Which statements about domestic electricity are correct ?

(1) All fuses should be installed in the live wire.
(2) An electrical appliance can work even without an earth wire.
(3) An electrical appliance having a fuse in it can protect people from getting an electric shock.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2012 Q33

33.



The figure shows a simple domestic circuit for an electric iron. The fuse will blow when which of the following points are short-circuited ?

(1) X and Y
(2) Y and Z
(3) X and Z

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2016 Q30

*30. A sinusoidal a.c. of a certain frequency delivers a r.m.s. voltage $V_{\text{r.m.s.}}$. If its frequency is doubled and its peak voltage is halved, what would be the r.m.s. voltage ?

A. $\frac{1}{2}V_{\text{r.m.s.}}$

B. $\frac{1}{\sqrt{2}}V_{\text{r.m.s.}}$

C. $\frac{1}{2\sqrt{2}}V_{\text{r.m.s.}}$

D. $V_{\text{r.m.s.}}$

2023 Q28

28. Which of the following statements about domestic circuits is/are correct ?

(1) The live (L) wire is sometimes positive and sometimes negative with respect to the neutral (N) wire.
(2) The kilowatt-hour meter measures the total power consumed by the domestic electrical appliances.
(3) With more electrical appliances connected, the total resistance of the domestic circuit becomes smaller.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

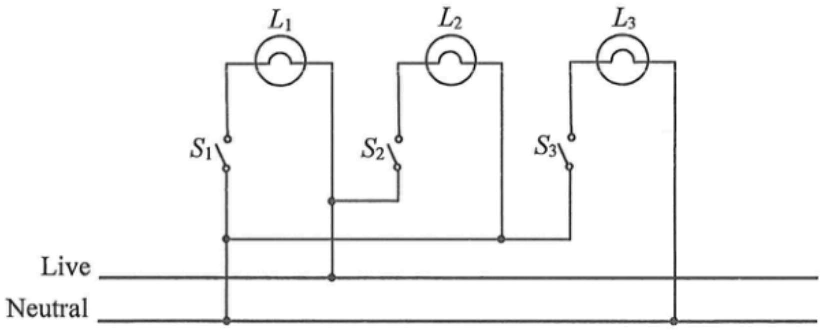
2024 Q30

*30. A power station generates 217 MW of power for transmission to downtown under a voltage of 132 kV via a transmission cable which has an effective resistance of 0.02 Ω km^{-1}. The power loss for a total cable length of 40 km is about

A. 2.16 MW.
B. 6.49 MW.
C. 13.0 MW.
D. 21.6 MW.

2025 Q27

27. The figure shows how lamps L_1, L_2 and L_3 are connected in the ring circuit of a house. There are some wrong connections in the circuit.



If switches S_2 and S_3 are closed while switch S_1 remains open, which of the lamps will light up ?

A. L_2 only
B. L_3 only
C. L_2 and L_3 only
D. none of them

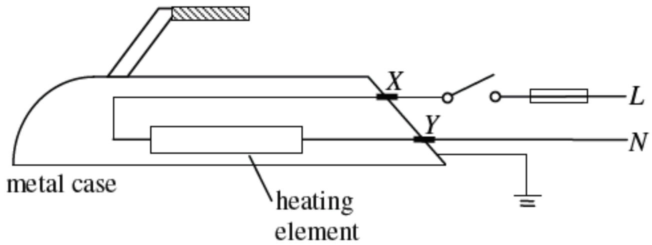
2026 Q25

25. In household appliances, the function of the earth wire is

A. to provide a return path for the current during normal operations.
B. to provide a return path for the current when the neutral wire is disconnected.
C. to break the circuit when the live wire touches the metal case.
D. to provide a low-resistance path when the live wire touches the metal case.

pp Q30

30.



The figure above shows the main parts of an electric iron. In which of the following situations will the fuse blow when the switch is closed ?

A. The heating element is broken and becomes an open circuit.
B. The earth wire is worn out and becomes disconnected.
C. The insulation at contact point *X* is worn out so that the wire touches the metal case.
D. The insulation at contact point *Y* is worn out so that the wire touches the metal case.

sap Q25

25. The table shows three electrical appliances which Clara used in a certain month :

Appliance	Rating	Duration
Air-conditioner	220 V, 1200 W	250 hours
television	220 V, 250 W	80 hours
computer	220 V, 150 W	60 hours

Calculate the cost of electricity used.
Note : 1 kW h of electricity costs $ 0.86.

A. $ 62.25
B. $ 73.79
C. $ 282.94
D. $ 536.64

sap Q26

26. If a 15 A fuse is installed in the plug of an electric kettle of rating '220 V, 900 W', state what happens when the kettle is plugged in and switched on.

A. The kettle will not operate.
B. The kettle will be short-circuited.
C. The output power of the kettle will be increased.
D. The chance of the kettle being damaged by an excessive current will be increase

sap Q33

*33. Power is transmitted over long distances at high alternating voltages. Which statements are correct ?

(1) Alternating voltages can be stepped up or down efficiently by transformers.
(2) For a given transmitted power, the current will be reduced if a high voltage is adopted.
(3) The power loss in the transmission cables will be reduced if a high voltage is adopted.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)